

Daiyang He

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Education

2023–Present **Sichuan University, SCUPI**
Industrial Engineering
B.S. (GPA: 3.85, 8/73)

- Research direction: **Mechanism-Informed AI for Clinical Insights**
- Keywords: AI4Science, Deep Learning, CFD, Physics Informed Machine Learning, System Engineering

Research Papers

- Y. Qiu*, **Daiyang He***, Z. Jiang, S. Cheng, Y. Tai, K. Li, H. Wu. *Open-source and fully automated multi-organ CT analysis for non-invasive portal hypertension evaluation*. *npj Digital Medicine*, Under Review, 2026.
- C. Weng*, **Daiyang He***, S. Cheng, D. Yuan, T. Wang, W. Zeng, Y. Luo, J. Wang, K. Li, Y. Qiu. *Noise-Pressure Constrained Liutex Method for Robust Vortex Identification in 4D Flow MRI of Abdominal Aortic Aneurysms*. *Computer Methods and Programs in Biomedicine*, Major Revision, 2026.
- S. Cheng, Y. Qiu, **Daiyang He**, M. L. Wang, C. Weng, D. Yuan, K. Li. *Helical vortex arising from aberrant right subclavian artery induces occurrence of dissection tears*. *Physics of Fluids*, 2025.
- S. Cheng, L. Liu, Y. Qiu, Y. Diao, **Daiyang He**, F. Yan, C. Wang, K. Li. *Hypoxia promotes blood blister-like aneurysms development: A CFD study*. *Journal of Biomechanics*, 2025.

Academic Competitions and Achievements

- 2025: **Undergraduate Training Program (National Level)**, "ShadowSift: An Efficient Real-Time Detection System for Low-Quality Deepfake Videos.", Sichuan University
- 2025: **Third Prize**, National Collegiate Mathematical Contest in Modeling, China Society for Industrial and Applied Mathematics.
- 2024: **Third Prize**, Southwest Division of the China Undergraduate Physics Tournament (CUPT), Chinese Physical Society.
- 2024: **S Award**, American Collegiate Mathematical Contest in Modeling, Consortium for Mathematics and its Applications.
- 2024–2026: **Excellent Student, Outstanding Students Leader, Comprehensive Scholarship**, Sichuan University.

Work Experience

2023–Present **Research Assistant, SCUPI BioFluid Lab**

- Conduct CT/MRI image analysis and CFD simulation.
- Develop the web-based automatic portal hypertension assessment pipeline **ALPHA** (<http://scupi.alpha.hxmedbot.com:10910/>).
- Explore physics-informed machine learning modeling for CFD surrogate.

2024–2026 **Student Organizations, SCUPI**

- **Leader** of Technology and Science Association (TSA)
- **Teaching Assistant** of IE major core courses (Productivity Analysis, Information System, Interdisciplinary Research)

Competences & Languages

Programming	Python, C/C++, MATLAB, SQL, $\LaTeX$
Scientific Knowledge	Medical Image Analysis, Machine Learning, Harness Engineering, CFD, Probability & Statistics, Mathematical Analysis
Software	Mimics, 3D Slicer, ANSYS Fluent, ICEM CFD, ParaView, TecPlot
Languages	English (fluent reading, writing, speaking), TOEFL 97